

Papers Sz Lyncis

Coen Beemster

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1 Variations in the period and in the light-curve of SZ Lyncis - van Genderen, A. M. (1967)

Dr. Arnout Martinus van Genderen is a Dutch Astronomer who is right now at the age of 90 years old still active in the field at Universiteit van Leiden. Back in 1967, van Genderen was the first one to recognise the sin-like properties of the O-C graph.

2 Photometric properties and evidence of duplicity for SZ Lyncis.- Barnes, T. G., III ; Moffett, T. J. (1975)

Barnes and Moffet where in 1975 the first ones to suggest that the sin-like properties of the O-C graph where caused by LTTE due to the fact that the cepheid could orbit another mass, a spectroscopic binary star system.

3 Mouvement orbital de la cepheide naine SZ LYN. - Bardin, Cl. ; Imbert, M. (1984)

Bardin and Imbert where in 1984 the ones to proof the theory of Barnes and Moffet, they proofed with measurements of its radial velocity that SZ Lyncis must orbit a mass (that we can't really see).

4 Orbital and Photometric Properties of SZ Lyncis - Moffett, T. J. ; Barnes, T G., III ; Fekel, F. C., Jr. ; Jefferys, W. H. ; Achtermann, J. M. (1988)

In 1988, Barnes and Moffet published (with some other researchers) another important paper on SZ Lyncis in which they managed to determine the orbital and the pulsation period very accurately, they then also determined the excentricity of the orbit.

5 SZ Lyn: New pulsational and orbital elements based on old and recent photometric observations - K.D. Gazeas, P.G. Niarchos, K.A. Boutsia (2004)

At the time when we where setting up this research, this was the paper where we thought that they measured the properties of SZ Lyncis where we where interested in the most accurately.

6 Observations and Orbital Analysis of the High-Amplitude Delta Scuti Star SZLyncis: The Unusual Orbital Precession - Li, Lin-Jia ; Qian, Sheng-Bang (2013)

This is one of the two papers in which the properties of SZ Lyncis are measured at the highest rate of accuracy.

7 Asteroseismology of SZ Lyn using multiband high time resolution photometry from ground and space - Adassuriya, J. ; Ganesh, S. ; Gutiérrez, J. L. ; Handler, G. ; Joshi, Santosh ; Jayaratne, K. P. S. C. ; Baliyan, K. S. (2021)

This is one of the two papers in which the properties of SZ Lyncis are measured at the highest rate of accuracy.

8 CoRoT-TESS eclipsing binaries with light-travel-time effect - T Hajdu , B Matécsa , J M Sallai , A Bódi (2022)

Samenvatting "CoRoT-TESS eclipsing binaries with light-travel-time effect" - Coen Beemster (2026)

Simulation (Download and open to use)

In this paper from 2022 the research relies on exactly the same phenomenon as in our research even the physical situation is slightly different. I've also made a summery (in Dutch) about this paper and made a simulation in which you can see the effects of the physical sistuation on the O-C graph.